

Venkata Rajesh Paidipala

EMAIL: pvr12116@gmail.com

PH: 513-394-3009

OH, Cincinnati (willing to relocate)

PROFESSIONAL SUMMARY:

Experienced Network Engineer with hands-on expertise in **routing, switching, and security**, including **BGP, OSPF, MPLS, VLANs, VTP, firewalls (Cisco ASA, Palo Alto), and VPN solutions** for secure and reliable networks.

- Skilled in **network automation and DevOps**, using **Python, Ansible, and Terraform** to streamline configurations, reduce manual work, and improve operational efficiency.
- Proficient in **cloud networking** on **AWS and Azure**, managing **VPC, EC2, IAM, and hybrid cloud integration** to enhance scalability and security.
- Strong troubleshooting skills with experience in **Wireshark, SNMP monitoring, and QoS policies** to optimize **network performance, security, and traffic management**.
- **Skilled in securing and configuring switches and firewalls**, with the flexibility to adapt to Fortinet security products as needed.
- Implemented and tested **disaster recovery solutions to minimize downtime** and ensure uninterrupted operations for **critical business functions**.
- A team player with a **problem-solving mindset**, eager to **collaborate on innovative networking projects** and contribute to building **efficient, secure, and scalable infrastructures**.

TECHNICAL SKILLS:

Networking:

- OSPF, IS-IS, BGP, RIP, EIGRP, RSVP-TE, MP-BGP, L3VPN, VRF, VLAN, VTP, QoS, IP/MPLS, Segment Routing, BRAS, DNS, DHCP, NTP, SNMP.
- security protocols (PLCs, SCADA), Industrial Ethernet (Layer 2/3)

Cloud & Virtualization:

- AWS (VPC, EC2, S3, Route53, IAM, CloudFormation), Azure Networking, GCP Networking, Hyper-V, VMware

Automation & DevOps:

- Python, Ansible, Terraform, Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions, GitLab CI)

Security:

- Firewalls (Cisco ASA, Palo Alto, FortiGate, Checkpoint), IDS/IPS, SIEAAA (TACACS+, RADIUS), Zero Trust

Tools & Protocols:

- NMS, EMS, Wireshark, PRTG, SolarWinds, NetFlow, Syslog, TCP/IP, SNMP, DHCP, DNS, Nmap, Metasploit, Cisco Packet Tracer, SonicWall, Netgear.

Scripting & Programming:

- Python, Bash, YAML, JSON, PowerShell

Version Control & Project Management:

- Git, Confluence, Agile & Scrum Methodologies

Operating Systems & Administration:

- Linux (CentOS, Ubuntu, Red Hat, Debian), Shell Scripting

Threat Prevention:

- IPS, DDoS/DoS Attack Detection, Vulnerability Scanning

Cybersecurity:

- IAM, Security Policies, Threat Analysis, Incident Response, SIEM

Experience:

Company: Tejas Networks

Sep 2022 – Dec 2023

Role: Research and Development PE Engineer

Environment & Skills:

- **MPLS, OSPF, BGP, EIGRP, RSVP-TE, L2VPN, VPLS, SD-WAN, LAN, WAN, DWDM, NMS, EMS, Python, Linux, Bash**
- Testing frameworks, Manual Testing, Automation Testing, Network Protocols

Responsibilities:

- Conducted **manual testing of security protocols (OSPF, BGP, EIGRP, RSVP-TE)**, ensuring secure network operations across multi-vendor environments.
- Tested **MPLS** protocols by evaluating security features and ensuring secure integration within service provider networks.
- Developed and validated test plans for **OSPFv3, L2VPN, and VPLS**, ensuring seamless network integration for enterprise and **SD-WAN** environments.
- Conducted testing of switches and routers through **NMS and EMS** to monitor performance, **configure devices**, and ensure compliance.
- Managed **VLANs, VTP and VPN technologies** (IPsec, OpenVPN), and firewall policies (Cisco ASA, Palo Alto) to establish secure communication channels.
- Integrated Tejas equipment into the core network by testing with **Nokia, Aruba, Juniper, and Cisco devices**, enhancing **service availability and network performance**.
- Exposure to automation tools like **Ansible and Git for scalable** configuration and version control (currently expanding usage).
- Part of the successful completion **VAPT (Vulnerability Acceptance and Penetration Testing)** Testing of **TJ1400P-H switch**.
- Utilized **Spirent and IXIA** for network testing and performance validation.
- Performed **DWDM testing on Tejas Networks TJ1600-11, TJ1600-6, and TJ1600-2** for Power Grid in a **500 km setup**, ensuring signal integrity, network reliability, and system compliance.

Company: Capgemini

Nov 2021 – August 2022

Role: Network Analyst

Environment & Skills:

IP Networking, Network Protocols, MPLS, QoS, Network Troubleshooting, Wireshark, PRTG, SolarWinds, Data Analysis, Python

Responsibilities:

- Optimized **network performance** by configuring **L2 switches, OSPF, BGP, and VLANs** while ensuring **compliance with security protocols**.
- Conducted **traffic analysis** using **Wireshark** and **hping**, identifying potential **DDoS and DoS** vulnerabilities and analyzing network behaviour under stress conditions.
- Implemented **QoS** to prioritize critical **business traffic**, ensuring **network efficiency and security**.
- Managed and enforced network security policies, utilizing **SNMP for continuous monitoring**, setting up alerts for abnormal traffic patterns, and **mitigating potential security incidents** in real time.
- Provided **real-time support for production environments**, quickly diagnosing and resolving **network-related issues**.
- Implemented security infrastructure solutions, including **firewall configurations** and **IDS/IPS** systems.
- Developed **PowerShell and Python scripts** for **network automation and monitoring**.

Education:

Master of Science in Information Technology

Jan 2024 – April 2025

University of Cincinnati

GPA : 3.93

Projects:

1. Title: Comprehensive Network Optimization and Secure SD-WAN Deployment.

Technologies: OSPF, BGP, VLANs, Cisco IOS, ACLs, MPLS, SD-WAN, VPN, QoS, Network Security, Wireshark, Network Monitoring Tools.

Description:

- Deployed **OSPF, RIP, EIGRP, and BGP** for **dynamic routing, redundancy**, and high network reliability.
- Utilized **VLANs** to segment **sensitive data**, enhancing performance and security.
- Implemented firewalls, **VPN encryption, ACLs**, and intrusion detection for **robust network security**.
- Designed and deployed **SD-WAN for secure branch office connectivity**, improving application performance and **reducing operational costs**.
- Applied application-aware routing to prioritize **VoIP and video** conferencing, ensuring seamless communication.
- Conducted network assessments and testing to ensure **high availability, low latency**, and **optimal user experience**.

2. Title: Zero Trust Security Architecture on AWS

Technologies Used: AWS IAM, VPC, Security Groups, NACLs, AWS WAF, AWS SSO, CloudTrail, GuardDuty, Terraform.

Description:

- Built a Zero Trust architecture in **AWS** focused on **identity verification, least-privilege access**, and segmented network controls.
- Configured **IAM policies**, roles, and **multi-factor authentication (MFA)** to enforce access restrictions across services.
- Designed **VPC networks** with **private/public subnets**, applying **Security Groups and Network ACLs** for strict traffic control.
- Deployed **AWS WAF** to protect applications from **OWASP Top 10 threats** and used **Shield for DDoS mitigation**.
- Enabled centralized logging with CloudTrail **and used GuardDuty for threat detection and** anomaly reporting.
- Integrated **AWS SSO** with an **external identity** provider to streamline and **secure user authentication**.
- **Automated infrastructure using Terraform** to ensure consistent deployment and **policy compliance**.

CERTIFICATIONS:

- Cisco Certified Network Associate (CCNA)
- Introduction to Cybersecurity
- CompTIA Security+ (**In Progress**)
- Microsoft Azure Fundamentals (**AZ-900**)